The Human Body, Part One: Structural Systems

The Skeleton, the Muscles, and the Skin — An Audio-Depth Guide

How to Listen to This Document

This guide is written to be understood by ear alone, with no picture required. To do that, anatomy uses a small, consistent vocabulary for describing where things are in relation to everything else, and it's worth learning these six words before anything else, because every location in this document will be built from them.

Superior means toward the head; inferior means toward the feet. Anterior means toward the front of the body; posterior means toward the back. Medial means toward the midline — the imaginary line running straight down the center of the body through the nose and navel; lateral means away from that midline, toward the sides. Proximal means closer to the trunk of the body; distal means farther from the trunk, out toward the hands and feet. Superficial means closer to the skin's surface; deep means farther inside, closer to the body's core.

All of this is described from what anatomists call the anatomical position: standing upright, facing forward, arms at the sides, palms facing forward. Keep that image — a person standing at attention, palms out — in mind as the fixed reference point for everything that follows, and the geography of the body will build logically, layer by layer, from the bones outward to the skin.

The Skeletal System

The skeleton is the body's scaffolding: two hundred and six bones in the adult body, connected at more than three hundred joints, that give the body its shape, protect its softer organs, anchor the muscles that create movement, store minerals like calcium and phosphorus, and — inside certain bones — manufacture blood cells. It divides cleanly into two regions: the axial skeleton, which forms the central vertical core of the body (skull, spine, ribcage), and the appendicular skeleton, which forms the limbs and the girdles that attach those limbs to the core (shoulders, arms, hands, hips, legs, feet).

The Axial Skeleton — The Central Core

Start at the very top. The skull sits at the superior end of the entire skeleton and is really two sets of bones fused together: the cranium, a rounded dome of eight bones that encases and protects the brain, and the facial skeleton, fourteen bones anterior and inferior to the cranium that form the eye sockets, nasal cavity, cheekbones, and upper and lower jaw. The only moving joint in the skull is the jaw joint, where the mandible — the lower jawbone, the only bone in the face that isn't fixed in place — hinges just anterior to each ear.

Below the skull, the vertebral column, or spine, runs down the posterior midline of the body from the base of the skull to the pelvis, and it is not a straight rod — it curves in an S-shape when viewed from the side, which is what lets it absorb shock and support upright weight efficiently. The spine is built from thirty-three individual bones called vertebrae, stacked and grouped into five regions from superior to inferior. The cervical

spine is the neck: seven vertebrae, the smallest and most mobile in the column, allowing the head to nod, tilt, and rotate. The very first cervical vertebra, directly under the skull, is called the atlas, because like the mythological figure it holds up the weight of the head; the second is the axis, which the atlas pivots around, letting the head turn side to side. Below the neck, the thoracic spine runs through the upper and mid back: twelve vertebrae, one for each pair of ribs that attaches to it posteriorly. Below that, the lumbar spine is the lower back: five vertebrae, the largest and thickest in the column since they bear the most body weight, which is also why lower-back strain is so common. Below the lumbar spine, the sacrum is a single triangular bone formed from five vertebrae that fused together, sitting posterior and central to the pelvis, wedged between the two hip bones like a keystone. And at the very inferior tip of the spine, below the sacrum, the coccyx — the tailbone — is a small triangle of three to five tiny fused vertebrae, the vestigial remnant of a tail, that anchors some pelvic floor muscles.

Anterior to the thoracic spine, and attached to it, is the ribcage, which forms a protective cage around the heart and lungs. There are twelve pairs of ribs, each pair attaching posteriorly to one thoracic vertebra and curving anterior and slightly inferior around the sides of the chest. The top seven pairs, called true ribs, connect directly at the front to the sternum — the breastbone, a flat vertical bone running down the center of the anterior chest — by way of a strip of cartilage. Ribs eight through ten, the false ribs, connect to the sternum indirectly, their cartilage joining the cartilage of rib seven above them rather than reaching the sternum directly. Ribs eleven and twelve, the floating ribs, don’t connect to the sternum at all — they attach only posteriorly to the spine and end freely in the muscle of the back and side. The sternum itself has three parts stacked superior to inferior: the manubrium at the top, where the collarbones attach; the body, the long central shaft where most of the true ribs attach; and the xiphoid process, a small pointer of cartilage at the very inferior tip, roughly at the point where the ribs stop meeting in the front — the same landmark used to position the hands for CPR.

The Appendicular Skeleton — Shoulders, Arms, and Hands

The upper limb attaches to the axial skeleton through the shoulder girdle, made of two bones on each side: the clavicle, or collarbone, a slender curved bone running horizontally from the top of the sternum laterally to the shoulder, easily felt just under the skin; and the scapula, or shoulder blade, a flat triangular bone that sits posterior to the ribcage, floating free against the back muscles rather than joining any other bone directly — it’s held in place entirely by muscle, which is why the shoulder has the greatest range of motion of any joint in the body.

From the lateral point of the scapula, the arm proper begins. The humerus is the single long bone of the upper arm, running from the shoulder joint proximally down to the elbow distally. At the elbow, the humerus meets two bones of the forearm side by side: the ulna, on the medial side (the pinky-finger side) when the palm faces forward, which forms the main hinge of the elbow; and the radius, on the lateral side (the thumb side), which rotates around the ulna to let the forearm twist — the motion of turning a doorknob or turning your palm up and down. At the wrist, these two forearm bones meet a cluster of eight small carpal bones, packed tightly in two rows of four, which form the flexible base of the hand. Distal to the carpals, five metacarpal bones fan out to form the palm, one leading to each finger. And each finger is built from phalanges — three per finger (proximal, middle, and distal), except the thumb, which has only two.

The Appendicular Skeleton — Hips, Legs, and Feet

The lower limb attaches through the pelvic girdle, which is different from the shoulder girdle in one important way: it’s built for weight-bearing stability rather than mobility,

so its bones are thick, heavy, and fused solid rather than loosely suspended. Each hip bone is actually three bones fused into one by adulthood — the ilium, the broad flaring wing you feel when you rest your hands on your hips, superior and lateral; the ischium, the lower posterior portion you sit on; and the pubis, the anterior portion, where the two hip bones meet each other at the midline in a joint called the pubic symphysis. Posteriorly, both hip bones attach to the sacrum, closing the pelvic ring into a solid basin that supports the entire upper body's weight and protects the pelvic organs.

From the lateral hip, where the hip bone forms a deep socket, the femur — the thigh bone — is the longest and strongest bone in the body, running from that hip socket down to the knee. At the knee, the patella, or kneecap, sits anterior to the joint, embedded in the tendon of the thigh muscle, acting as a protective shield and a mechanical lever. Below the knee, again there are two bones side by side: the tibia, medial and much larger, the true weight-bearing shin bone that you can feel as a hard ridge down the front of your lower leg; and the fibula, lateral and thinner, which doesn't bear much weight but provides an attachment point for muscles and stabilizes the ankle. At the ankle, the lower leg bones meet seven tarsal bones, the largest of which, the calcaneus, forms the heel. Distal to the tarsals, five metatarsal bones form the arch and ball of the foot, and each toe is built from phalanges just like the fingers — three per toe except the big toe, which has two.

Joints — Where Bones Meet and Move

A joint is simply any point where two bones meet, and different joints allow different kinds of movement based on their shape. Hinge joints, like the elbow and knee, move in one plane only, like a door — bending and straightening. Ball-and-socket joints, like the shoulder and hip, have a rounded head fitting into a cup-shaped socket, allowing movement in almost every direction, including full rotation — the trade-off being that the more mobile the joint, the less inherently stable it is, which is why shoulders dislocate far more often than hips. Pivot joints, like the one between the atlas and axis in the neck, allow rotation around a single axis, like a key turning in a lock. And gliding joints, like those between the carpal bones of the wrist, allow small sliding movements in multiple directions.

The Muscular System

Where the skeleton is the frame, the muscular system is what moves it. There are more than six hundred muscles in the human body, and they come in three distinct types. Skeletal muscle is the kind attached to bone and under voluntary, conscious control — this is what most of this section covers. Smooth muscle lines the walls of internal organs like the stomach, intestines, and blood vessels, working automatically and involuntarily to push contents along or adjust vessel diameter. Cardiac muscle exists only in the heart, a specialized involuntary muscle built to contract rhythmically for an entire lifetime without fatigue.

Skeletal muscles work by attaching to bone at two ends through tendons — tough, rope-like connective tissue. The end that stays relatively fixed during a movement is called the origin, and the end that moves is called the insertion. Muscles can only pull, never push, which is why they nearly always work in opposing pairs: when the biceps on the front of the arm contracts to bend the elbow, the triceps on the back of the arm must relax, and to straighten the elbow back out, the roles reverse. This pairing principle — one muscle as the "agonist" doing the work, its partner as the "antagonist" relaxing to allow it — holds true across nearly every joint in the body.

Muscles of the Head and Neck

The muscles of facial expression are unusual: unlike most skeletal muscles, many of them attach not from bone to bone but from bone into the skin itself, which is exactly why they can wrinkle the skin around the eyes, lift the corners of the mouth, or raise the eyebrows. The muscles of mastication — chewing — are more conventional. The masseter, one of the strongest muscles in the body relative to its size, runs from the cheekbone down to the angle of the jaw and is easily felt by clenching the teeth and touching the back of the jaw just anterior to the ear. The temporalis is a broad fan-shaped muscle covering the side of the skull above and anterior to the ear, also closing the jaw.

In the neck, the sternocleidomastoid is a prominent muscle running diagonally from just behind the ear down to the sternum and collarbone, visible and easily felt when the head turns to the opposite side — it’s the muscle that stands out in relief when someone strains to look sharply over one shoulder. The trapezius, though it’s really a back-and-shoulder muscle, has its most superior fibers reaching up into the neck and skull, and this large kite-shaped muscle spans from the base of the skull, out to both shoulders, and down the length of the upper and mid spine, giving it a role in shrugging the shoulders, tilting the head, and stabilizing the shoulder blades all at once.

Muscles of the Trunk

On the anterior chest, the pectoralis major is the large fan-shaped muscle that gives the chest its bulk, originating along the sternum and collarbone and inserting onto the humerus, responsible for pulling the arm across the body and toward the trunk. On the posterior side, the latissimus dorsi — often just called the “lats” — is the broadest muscle in the body, a wide sheet covering most of the lower and mid back, originating from the lower spine and pelvis and sweeping up and laterally to insert on the humerus, pulling the arm down and back, the muscle most engaged in a pull-up or a rowing motion. Between the shoulder blades, the rhomboids run from the spine out to the medial edge of each scapula, pulling the shoulder blades together — the muscles you feel working when you squeeze your shoulder blades back.

Running the entire length of the spine, posteriorly, on either side of the vertebrae, the erector spinae is actually a group of several long muscles that work together to extend the spine and hold the body upright against gravity — this is the muscle mass you can feel as two firm columns running up either side of the backbone.

On the anterior abdomen, the rectus abdominis runs vertically from the ribcage down to the pubic bone, the muscle responsible for the “six-pack” appearance when well-developed, since it’s divided into segments by bands of connective tissue. Lateral to it, the external and internal obliques run diagonally across the sides of the abdomen, enabling twisting and side-bending, and deep to both of these, the transverse abdominis wraps around the trunk horizontally like a natural corset, the deepest of the abdominal muscles, providing core stability. And separating the chest cavity from the abdominal cavity entirely is the diaphragm, a dome-shaped sheet of muscle that arches up into the chest at rest; when it contracts, it flattens and drops inferiorly, expanding the chest cavity and drawing air into the lungs — this is the primary muscle of breathing, and it sits directly superior to the stomach and liver, which is part of why a full stomach can make deep breathing feel more difficult.

Muscles of the Upper Limb

Capping the shoulder, the deltoid is the rounded muscle that gives the shoulder its shape, originating from the collarbone and shoulder blade and wrapping over the top of the shoulder joint to insert on the humerus, responsible for lifting the arm out to the side and forward. On the anterior upper arm, the biceps brachii — commonly just “the biceps” — runs from the shoulder blade down to the forearm bones, flexing the elbow

and rotating the forearm to turn the palm upward. Directly posterior to it, on the back of the upper arm, the triceps brachii, as its name suggests, has three heads of origin converging into one muscle that inserts at the elbow, and it's the primary muscle that straightens the arm. In the forearm, two opposing muscle groups run along the anterior and posterior surfaces: the flexors, on the anterior (palm-side) surface, curl the wrist and fingers, while the extensors, on the posterior surface, straighten them back out — both groups' tendons cross the wrist and anchor onto the bones of the hand and fingers, which is why so much hand and finger movement is actually powered by muscle bulk located up in the forearm rather than in the hand itself.

Muscles of the Lower Limb

At the hip, the gluteal muscles form the buttocks: the gluteus maximus, the largest and most superficial, is the main muscle that extends the hip — powering the drive in walking uphill or standing up from a seated position — while the gluteus medius and gluteus minimus lie deeper and more superior and lateral, primarily stabilizing the pelvis during walking so it doesn't dip with every step.

On the anterior thigh, the quadriceps femoris is a group of four muscles — the rectus femoris running down the center, and three vastus muscles wrapping around the sides and deep to it — that converge into a single tendon crossing the kneecap and inserting just below the knee, making this group responsible for straightening the knee, the primary muscle used in kicking or standing up from a squat. On the posterior thigh, directly opposing the quadriceps, the hamstrings are a group of three muscles running from the pelvis down to just below the knee, responsible for bending the knee and extending the hip — the muscle group most commonly strained in sprinting.

Below the knee, on the posterior calf, the gastrocnemius is the muscle that gives the calf its visible diamond shape, running from just above the knee down to the heel by way of the Achilles tendon — the thickest and strongest tendon in the body. Deep to the gastrocnemius, the soleus is a flatter muscle that also inserts via the Achilles tendon, and together these two muscles are what point the foot downward, as in pushing off the ground while walking or standing on tiptoe. On the anterior side of the lower leg, the tibialis anterior runs alongside the shin bone and pulls the foot upward, the muscle that fatigues and cramps as "shin splints" in new runners.

The Integumentary System — Skin, Hair, and Nails

The integumentary system is the body's outermost boundary and, by surface area, its largest organ system — the skin alone covers roughly twenty square feet in an average adult. It has three layers, superficial to deep. The epidermis is the thin, tough outer layer, made mostly of dead, flattened, keratin-hardened cells that are constantly shed and replaced from below — it's what you actually touch when you touch skin, and it contains no blood vessels of its own, which is why a light scrape that only removes epidermis doesn't bleed. Deep to the epidermis, the dermis is a much thicker layer of dense connective tissue that houses the structures that make skin functional: blood vessels, nerve endings for touch, pressure, temperature, and pain, hair follicles, sweat glands, and oil-producing sebaceous glands. Deepest of all, the hypodermis, or subcutaneous layer, is not technically part of the skin itself but lies immediately beneath the dermis — a layer of fat and connective tissue that insulates the body, cushions it against impact, and stores energy.

Functionally, skin is the body's primary barrier against the outside world, keeping pathogens, chemicals, and excess water out while keeping the body's own fluids in. It regulates temperature by adjusting blood flow near the surface and by sweating to cool

through evaporation. It’s the body’s largest sensory organ, reporting touch, temperature, and pain from every point on its surface back to the nervous system. And when exposed to sunlight, it synthesizes vitamin D, a hormone essential for calcium absorption and bone health. Hair and nails are both specialized extensions of the epidermis, built from the same tough keratin protein: hair grows from follicles rooted in the dermis, and nails grow from a matrix at their base, both essentially “dead” structures by the time they’re visible, with no nerve endings of their own — which is why cutting hair or trimmed nail tips causes no pain.

Putting the Structural Layers Together

Taken together, these three systems form the body’s outer architecture from the inside out. At the center is the skeleton — the two-hundred-and-six-bone frame, its axial core running skull to tailbone down the midline, its appendicular limbs branching out from shoulders and hips. Wrapped around that frame, muscle by muscle, is the muscular system, six hundred-plus muscles anchored to bone by tendon, working in opposing pairs to pull the skeleton into every posture and movement the body makes. And wrapped around all of it, as the outermost boundary between the body and the world, is the skin, with its three layers running from the tough dead cells at the surface down through the living, sensing dermis to the insulating fat beneath. Every organ system described in the documents that follow — nervous, cardiovascular, respiratory, digestive, and the rest — lives inside this same frame, packed into the spaces the skeleton encloses and the muscles move.